Practice Exam 3

MTH 207 (Fall 2026) — J. Baggett

This is practice, not a preview. The real exam will **not** be these problems with the numbers changed. Expect different functions, and expect at least one technique this sheet happens not to show. Everything on the review list is fair game whether or not it turns up here — use this to find out what you cannot yet do, not to predict what will be asked.

This sheet is also deliberately **longer** than the real exam (128 points against 100) so that it covers more ground, so do not use it to judge timing. The real exam is 55 minutes.

First Name	Last Name

Before exam begins:

- Write your first name and last name above.
- Turn off all electronics and keep them out of sight: no cellular phones, not even to tell time (and not even if it's just in airplane mode). No iPods, wearing of headphones, etc.

As soon as exam starts:

- Take a quick breath to relax!!
- Check that you have all 7 pages of the exam. (The number of pages includes this cover page.)

During the exam:

- **TO EARN POINTS YOU MUST SHOW YOUR WORK:**
  - Explain your reasons and justify your steps.
  - Demonstrate that you understand the *process*.
  - Present your logic in an organized manner on paper.
- Keep your eyes on your own exam!
- No calculators, phones, or other electronics at all. They should be turned off!
- No notes, books, or other outside resources.
- When you are told to stop, you must stop **IMMEDIATELY**. This is in fairness to all students. Do not think that you are the exception to this rule. Continuing to work on the exam after time has been called will have serious consequences.

(3pts ea.) **1.** Determine whether each statement is **true** or **false**. If it is true, say why in one sentence. If it is false, explain why or give an example that disproves it.

(a) There exists a function f such that $f(x) < 0$, $f'(x) < 0$, and $f''(x) > 0$ for all x.

Solution:

FALSE. Concave up and decreasing forces f upward eventually; staying negative forever is impossible.

(b) In an optimization problem on a closed interval, the endpoints must be checked even when the derivative is never zero there.

Solution:

TRUE. The Extreme Value Theorem guarantees a maximum and a minimum, and if no interior critical point provides one, an endpoint must.

(c) If $\lim_{x\to a} \dfrac{f'(x)}{g'(x)}$ does not exist, then $\lim_{x\to a} \dfrac{f(x)}{g(x)}$ does not exist either.

Solution:

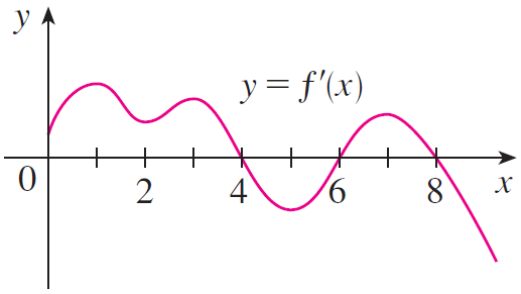
FALSE. l'Hopital's Rule runs one way only: if the limit of the quotient of derivatives exists, so does the original. Failure of the first says nothing about the second.

(d) $\displaystyle\int_{-1}^{1} \left(x^5 - 6x^9 + \frac{\sin x}{(1+x^4)^2} \right) dx = 0$

Solution:

TRUE. Every term is odd and the interval is symmetric about 0, so the signed areas cancel.

(3pts ea.) **2.** The graph of the first derivative f' of a function f is shown.



(a) On what intervals is f increasing? Explain briefly.

(b) At what values of x does f have a local maximum? Explain briefly.

(c) On what intervals is f concave upward? Explain briefly.

(d) What are the x-coordinates of the inflection points of f? Why?

(5^{pts}) **3.** Draw the graph of a function that satisfies $f'(x) < 0$ and $f''(x) < 0$ for all x.

Solution:
Two conditions, and they are independent: $f' < 0$ means the curve is **falling** everywhere; $f'' < 0$ means it is **concave down** everywhere — the slope is itself getting more negative, so the fall steepens as x increases.
So the graph drops from upper left to lower right and bends *downward*, like the right-hand half of a dome tipping over. Any of these works:

$$y = -x^2 \ \ (x > 0), \qquad y = -e^x, \qquad y = \ln\left(\frac{1}{x}\right) - x$$

Check $y = -e^x$: $y' = -e^x < 0$ and $y'' = -e^x < 0$ for every x.
A common wrong answer is a curve that falls but flattens out (like $y = 1/x$ for $x > 0$) — that is concave *up*, since the slope rises toward 0.

(5^{pts} ea.) **4.** Evaluate each limit or state why it doesn't exist. As a reminder, justify your steps.

(a) $\lim_{x\to\infty} \dfrac{x^2}{e^x}$

(b) $\lim_{x\to\infty} (1+\sin 3x)^{1/x}$

(c) $\lim_{x\to 0^+} x \ln x$

5. On this page, let $f(x) = \dfrac{e^x}{x+2}$. Then $f'(x) = \dfrac{(x+1)e^x}{(x+2)^2}$ and $f''(x) = \dfrac{(x^2+2x+2)e^x}{(x+2)^3}$.
Note: the expressions e^x and $x^2+2x+2 = (x+1)^2+1$ are **both** always positive for all x.

(a) *(5 pts)* Which x-values are critical numbers?

(b) *(7 pts)* On which x-intervals is f increasing? On which x-intervals is f decreasing?

(c) *(8 pts)* On which x-intervals is f concave up? On which x-intervals is f concave down?

Solution:
(a) Critical numbers. f itself is undefined at $x=-2$, so -2 cannot be a critical number — critical numbers must lie in the domain of f. Elsewhere $f'(x) = \dfrac{(x+1)e^x}{(x+2)^2} = 0$ requires $(x+1)e^x = 0$, and e^x is never zero. So the only critical number is $\mathbf{x = -1}$.
(b) Increasing / decreasing. Since $e^x > 0$ and $(x+2)^2 > 0$ for $x \neq -2$, the sign of f' is exactly the sign of $x+1$:

$$f' < 0 \text{ on } (-\infty,-2) \text{ and } (-2,-1), \qquad f' > 0 \text{ on } (-1,\infty)$$

So f is **decreasing on** $(-\infty,-2)$ **and on** $(-2,-1)$, and **increasing on** $(-1,\infty)$. The two decreasing intervals are listed separately because f does not exist at -2.
(c) Concavity. The note says $e^x > 0$ and $x^2+2x+2 > 0$ always, so the sign of f'' is the sign of $(x+2)^3$ — which is the sign of $x+2$:

$$f'' < 0 \text{ on } (-\infty,-2), \qquad f'' > 0 \text{ on } (-2,\infty)$$

Concave down on $(-\infty,-2)$**; concave up on** $(-2,\infty)$**.** There is no inflection point, because f is not defined at -2.

(20pts) **6.** A running track 400 meters long (around the outside) is to be constructed with two parallel straightaways (line segments) of length s joining two semicircular curves of radius r. What are the *dimensions of the track* such that the area of the rectangular part of the enclosed field is maximized? Use units and give mathematical justification to show you've found the absolute maximum.

Solution:

Let s be the length of each straightaway and r the radius of each semicircular end. The two semicircles together form one full circle, so the 400 m perimeter gives the **constraint**

$$2s + 2\pi r = 400 \quad \Longrightarrow \quad s = 200 - \pi r$$

The rectangular field has length s and width $2r$, so the **objective** is

$$A = 2rs = 2r\,(200 - \pi r) = 400r - 2\pi r^2$$

Maximize over $0 < r < \dfrac{200}{\pi}$:

$$\frac{dA}{dr} = 400 - 4\pi r = 0 \quad \Longrightarrow \quad r = \frac{100}{\pi}$$

$\dfrac{d^2A}{dr^2} = -4\pi < 0$, so this critical number is a maximum.

Then $s = 200 - \pi\left(\dfrac{100}{\pi}\right) = 100$.

Dimensions: straightaways $s = 100$ m, radius $r = \dfrac{100}{\pi} \approx 31.8$ m — so the rectangle is 100 m by $\dfrac{200}{\pi} \approx 63.7$ m, with area $\dfrac{20000}{\pi} \approx 6366$ m^2.

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(6pts) **7.** Evaluate $\displaystyle\int_{-5}^{5} \left(x - \sqrt{25 - x^2}\right) dx$

Solution:

Split the integral and read each piece as an area or a symmetry.

$$\int_{-5}^{5} \left(x - \sqrt{25 - x^2}\right) dx = \int_{-5}^{5} x\,dx \; - \; \int_{-5}^{5} \sqrt{25 - x^2}\,dx$$

First piece: x is an **odd** function and the interval is symmetric about 0, so the signed areas cancel exactly: $\displaystyle\int_{-5}^{5} x\,dx = 0$.

Second piece: $y = \sqrt{25 - x^2}$ is the *upper half* of the circle $x^2 + y^2 = 25$ (radius 5). Over $[-5, 5]$ that is the whole semicircle, whose area is

$$\frac{1}{2}\pi(5)^2 = \frac{25\pi}{2}$$

Therefore

$$\int_{-5}^{5} \left(x - \sqrt{25 - x^2}\right) dx = 0 - \frac{25\pi}{2} = -\frac{\mathbf{25\pi}}{\mathbf{2}} \approx -39.3$$

No antiderivative is needed for either piece.

(16pts) **8.** Determine the definite integral $\int_0^1 3x\,dx$ as a limit of Riemann sums. (You may use any sampling rule.) You must use the method of Riemann sums: no points will be awarded if you do not set up a Riemann sum, though you may optionally use another method to check your work. Some helpful sums are listed at the bottom.

Solution:

Divide $[0,1]$ into n equal pieces, so $\Delta x = \dfrac{1-0}{n} = \dfrac{1}{n}$, and take **right** endpoints $x_i = 0 + i\Delta x = \dfrac{i}{n}$.

$$\int_0^1 3x\,dx = \lim_{n\to\infty}\sum_{i=1}^{n} f(x_i)\,\Delta x = \lim_{n\to\infty}\sum_{i=1}^{n} 3\left(\frac{i}{n}\right)\frac{1}{n} = \lim_{n\to\infty}\frac{3}{n^2}\sum_{i=1}^{n} i$$

Using $\displaystyle\sum_{i=1}^{n} i = \frac{n(n+1)}{2}$,

$$= \lim_{n\to\infty}\frac{3}{n^2}\cdot\frac{n(n+1)}{2} = \lim_{n\to\infty}\frac{3(n+1)}{2n} = \lim_{n\to\infty}\frac{3}{2}\left(1+\frac{1}{n}\right) = \mathbf{\frac{3}{2}}$$

Check (not a substitute for the work): the region is a triangle with base 1 and height 3, area $\frac{1}{2}(1)(3) = \frac{3}{2}$. ✓

$$\sum_{i=1}^{n} i = \frac{n(n+1)}{2}, \qquad \sum_{i=1}^{n} i^2 = \frac{n(n+1)(2n+1)}{6}, \qquad \sum_{i=1}^{n} i^3 = \left(\frac{n(n+1)}{2}\right)^2$$

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(12pts) **9.** Estimate the area under the graph of $f(x) = \dfrac{1}{x}$ from $x = 1$ to $x = 3$ using four rectangles of equal width and **left** endpoints. Is your estimate larger or smaller than the true area? Explain how you know.

Solution:

$\Delta x = \dfrac{3-1}{4} = \dfrac{1}{2}$, and the **left** endpoints of the four subintervals are $x = 1,\ 1.5,\ 2,\ 2.5$.

$$L_4 = \Delta x\Big[f(1) + f(1.5) + f(2) + f(2.5)\Big] = \frac{1}{2}\left[1 + \frac{2}{3} + \frac{1}{2} + \frac{2}{5}\right]$$

$$= \frac{1}{2}\cdot\frac{77}{30} = \frac{77}{60} \approx \mathbf{1.283}$$

Larger or smaller? Larger. On $[1,3]$ the function $f(x) = \dfrac{1}{x}$ is **decreasing**, so on each subinterval the left endpoint is where f is *biggest*. Every rectangle therefore pokes above the curve, and the sum overestimates.

(For comparison the true value is $\ln 3 \approx 1.099$ — but the reasoning, not the comparison, is what earns the points.)

(10pts) **10.** Suppose $\int_0^5 f(x)\,dx = 10$ and $\int_0^5 g(x)\,dx = -3$. Evaluate each of the following, or say why it cannot be determined.

(a) (5 pts) $\displaystyle\int_0^5 \big(2f(x) - 3g(x)\big)\,dx$

(b) (5 pts) $\displaystyle\int_5^0 f(x)\,dx$

Solution:
(a) Linearity — constants come out and sums split:

$$\int_0^5 \big(2f(x) - 3g(x)\big)dx = 2\int_0^5 f(x)\,dx - 3\int_0^5 g(x)\,dx = 2(10) - 3(-3) = 20 + 9 = \mathbf{29}$$

Watch the double negative on the g term; $-3 \times -3 = +9$.
(b) Reversing the limits flips the sign:

$$\int_5^0 f(x)\,dx = -\int_0^5 f(x)\,dx = \mathbf{-10}$$

Both are determined — no extra information about f or g is needed, because both rules follow from the definition of the definite integral.

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